

Day 5 - MySQL Aggregate Functions Interview Guide

Scenario 1: 1. Employee Salary Summary

Scenario: HR wants the total monthly salary expense.

Interview Question: Write a MySQL query for the above scenario.

MySQL Query:

```
SELECT SUM(Salary) AS Total_Salary_Expense
FROM Employee;
```

Explanation: SUM() calculates the total salary paid to all employees.

Scenario 2: 2. Student Average Marks

Scenario: Find the average marks of students enrolled in the SQL course.

Interview Question: Write a MySQL query for the above scenario.

MySQL Query:

```
SELECT AVG(Marks) AS Average_Marks
FROM Student
WHERE Course='SQL';
```

Explanation: AVG() returns the average of the Marks column.

Scenario 3: 3. Product Price Analysis

Scenario: Find the highest and lowest product prices.

Interview Question: Write a MySQL query for the above scenario.

MySQL Query:

```
SELECT MAX(Price) AS Highest_Price,
       MIN(Price) AS Lowest_Price
FROM Product;
```

Explanation: MAX() returns the highest value and MIN() returns the lowest value.

Scenario 4: 4. Customer Registration Report

Scenario: Count the total number of customers.

Interview Question: Write a MySQL query for the above scenario.

MySQL Query:

```
SELECT COUNT(*) AS Total_Customers
FROM Customer;
```

Explanation: COUNT(*) counts all rows in the table.

Scenario 5: 5. Company Salary Dashboard

Scenario: Display employee count, total salary, average salary, highest salary and lowest salary.

Interview Question: Write a MySQL query for the above scenario.

MySQL Query:

```
SELECT COUNT(*) AS Total_Employees,
       SUM(Salary) AS Total_Salary,
       AVG(Salary) AS Average_Salary,
       MAX(Salary) AS Highest_Salary,
       MIN(Salary) AS Lowest_Salary
FROM Employee;
```

Explanation: Combines multiple aggregate functions into a single dashboard query.